AI-ML Virtual Internship

AI-Powered Product Image Search & Classification App using TensorFlow Lite

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PROGRAM: B.Tech CSE (AIML)
HOST ORGANIZATION: EduSkills Foundation (AICTE), supported by Google for Developers

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What This Presentation Covers

- Introduction — Internship overview & curriculum structure
- Organization Profile — AICTE, EduSkills Foundation & Google for Developers
- Problem Statement — Why visual product search is needed
- Objectives & Scope — Goals and boundaries of the project
- Technologies Used — TensorFlow, TFLite, Model Maker and toolchain
- System Architecture — How the app is structured
- Methodology — Week-wise plan & step-wise pipeline
- Outcomes & Results — Model performance and takeaways

Introduction — An 8-Week Journey into Applied ML

- 10-week… no — 8-week AI-ML Virtual Internship (April–June 2026), conducted under AICTE and the National Internship Portal, supported by the India Edu Program and Google for Developers.
- The program built a progressive understanding of applied machine learning — from programming neural networks with TensorFlow to on-device object detection, product image search and custom image classification.
- Learnings were consolidated into a single project: an AI-Powered Product Image Search & Classification App, directly mapping to the eight weekly modules.
- Completed with an overall grade of A (Very Good).

Organization Profile

- EduSkills Foundation
- Section-8 not-for-profit bridging academia & industry through virtual internships, in partnership with AICTE.
- Certified ISO 9001:2015 & ISO 21001:2018 — recognized as a Great Place to Work.

- Google for Developers / India Edu Program
- Google's global developer program providing free ML learning paths, codelabs and TensorFlow content.
- Supported the AI-ML curriculum with hands-on, certification-ready labs.

Problem Statement — Text Search Doesn't Fit Visual Needs

- Typing exact product names is slow and fails when the user doesn't know what the product is called.
- Cloud inference for every query is expensive, adds latency, and sends user images off-device — a privacy concern.

THE NEED

- A simple, fast, privacy-friendly system that detects and classifies products directly on the mobile device from a live camera feed, and visually matches them against a product catalogue.

Project Objectives — Six Goals Guiding the Build

- Program and train neural networks from first principles using TensorFlow, up to CNNs for image recognition.
- Implement object detection using pretrained detectors integrated into a mobile app.
- Train a custom object-detection model using TensorFlow Lite and Model Maker.
- Build a product image search feature using on-device object detection on static images and live camera feeds.
- Train a custom image classifier with transfer learning and integrate it into the app.
- Keep the solution simple and understandable, while touching every stage of the curriculum.

Scope of the Project — What's In and What's Out

• IN SCOPE
• Training a custom image classifier on a small sample product dataset (transfer learning).
• On-device object detection and classification with TensorFlow Lite.
• Visual matching against a small reference product catalogue.
• Prototype Android app built and demonstrated during the internship.

• OUT OF SCOPE
• ✗ Full-scale, production-grade deployment.
• ✗ Large real-world catalogues beyond the demo set.
• ✗ Enterprise backend, security or scaling infrastructure.
• ✗ Integration with live e-commerce systems.

Technologies & Tools — The TensorFlow-Powered Toolchain

TensorFlow / Keras	Train neural-network & classification models
TensorFlow Lite	On-device inference in the mobile app
TFLite Model Maker	Custom object detection via transfer learning
Google Colab + Python	Training, evaluation & scripting
Android Studio	App front-end, camera & result display
EfficientDet-Lite / MobileNet	Fast pretrained detection & classification bases

System Architecture — A Simple, Linear On-Device Pipeline



A camera frame is localized by a TFLite object detector; each region is classified by a custom TFLite image classifier; the predicted category and visual features are matched against the app's product catalogue, and ranked results are displayed — all inference on-device, fast and private.

Methodology — Week-Wise Structure of the Internship

Neural Networks & Object Detection (Weeks 1–3)

- W1: Program neural networks with TensorFlow — first principles to CNNs.
- W2: Get started with object detection — pretrained detectors in mobile apps.
- W3: Go further — custom detection with TFLite & Model Maker.

Product Image Search (Weeks 4–6)

- W4–5: On-device product image search — static images & live camera feeds.
- W6: Backend integration for the search app.

Image Classification (Weeks 7–8)

- W7: Custom image-classification models.
- W8: Integrate the custom classifier into the app.
- Final: Credential Validation & Grade Assessment.

The Project — A Six-Step Pipeline

1. Collect product-image dataset
2. Train classifier (transfer learning)
3. Convert model to TFLite
4. On-device object detection
5. Classify & match catalogue
6. Display ranked results

Weekly assessments tracked progress; the final grade combined weekly work, project submissions and the final assessment.

Outcomes & Results — Model Performance and Takeaways

93.5%	0.92	0.91	<150 ms
Classification Accuracy	Precision	Recall / F1-Score	On-device Inference Time

- End-to-end applied ML experience — data prep, training, TFLite conversion and on-device deployment.
- Practical skills: transfer learning, Model Maker, mobile camera integration.
- A working product image search & classification app prototype.
- Awareness of on-device AI benefits: low latency, offline use, user privacy.
- Completed the AICTE – EduSkills AI-ML Virtual Internship with grade A (Very Good).

Successfully Completed

AICTE – EduSkills AI-ML Virtual Internship

Overall Grade: A — Very Good
Certificate ID: 478b80131b258f6e5f86
Duration: April – June 2026 (8 Weeks)

Thank You

